

Convergence Investigation Report — case-c

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Investigation Identities

Original: **caseC_original**
Candidate: **caseC_candidate**

Linkage Strength-of-Evidence

50.4/100 — MODERATE

This score represents strength of evidence, not probability of guilt or identity certainty.

Signal Breakdown

Signal	Similarity	Contribution	Status	Evidence ID
temporal	0.798	+16.99	SUPPORTING	E-caseC_original-caseC_candidate-temporal
account	0.585	+9.35	SUPPORTING	E-caseC_original-caseC_candidate-account
infrastructure	0.050	-19.38	CONTRADICTING	E-caseC_original-caseC_candidate-infrastructure
activity	0.998	+19.27	SUPPORTING	E-caseC_original-caseC_candidate-activity
interaction_network	0.808	+13.35	SUPPORTING	E-caseC_original-caseC_candidate-interaction_network
linguistic	0.815	+10.83	SUPPORTING	E-caseC_original-caseC_candidate-linguistic

Supporting Evidence

temporal: {'matching_activity_hours': [8, 9, 10, 11, 13, 19, 20, 21, 22], 'matching_activity_windows': ['08:00–12:00', '13:00–14:00', '19:00–23:00'], 'observation_period_days': 14, 'similarity': 0.798}

account: {'creation_gap_days': 21.0, 'posting_interval_diff_minutes': 55.0, 'creation_decay_days': 60, 'posting_decay_minutes': 120, 'similarity': 0.585}

activity: {'identity_a_actions': {'login': 1, 'post': 7, 'upload': 2, 'message': 3, 'withdraw': 7, 'search': 2, 'browse': 1}, 'identity_b_actions': {'login': 1, 'post': 6, 'upload': 2, 'message': 3, 'withdraw': 6, 'search': 2, 'browse': 1}, 'strongest_matching_behaviors': ['post', 'withdraw', 'message'], 'similarity': 0.998}

interaction_network: {'identity_a_interaction_vector': [10.447, 7.791, 6.85, 4.289, 5.695], 'identity_b_interaction_vector': [3.419, 9.854, 3.748, 7.888, 1.583], 'shared_counterparties': [], 'shared_counterparty_count': 0, 'similarity': 0.808, 'note': 'Behavioral interaction patterns are attribution evidence; privacy/anonymity tools are excluded from this signal.'}

linguistic: {'similarity': 0.815, 'features': ['sentence length', 'punctuation pattern', 'function-word usage', 'vocabulary patterns', 'n-gram distribution'], 'representation': 'Synthetic stylometric representation', 'note': 'Prototype stylometry is synthetic; the similarity computation itself is cosine similarity.'}

Contradicting Evidence

infrastructure: {'identity_a_infra': 'infra_C_A', 'identity_b_infra': 'infra_C_B', 'matched_components': [], 'match_ratio': 0.0, 'match_type': 'none', 'identity_a_components': {'tls': 'tls_C1', 'service': 'svc_C1', 'certificate': 'cert_C1', 'server': 'srv_C1', 'asn': 'asn_C1'}, 'identity_b_components': {'tls': 'tls_C2', 'service': 'svc_C2', 'certificate': 'cert_C2', 'server': 'srv_C2', 'asn': 'asn_C2'}, 'similarity': 0.05, 'ps26151_infrastructure_evidence': {'ssl_certificates': [], 'certificate_domain_links': [], 'clearnet_domains': [], 'service_banners': [], 'exposed_server_status': [], 'descriptor_inconsistencies': []}, 'observations': [], 'note': 'Infrastructure score uses TLS/service/certificate/server/ASN component overlap; PS-specific clearnet/certificate/server-status/descriptor objects are explicitly represented as corroborating evidence.'}

Insufficient Evidence

None identified.

Threat-Actor Intelligence Layer

Shared objects: **1**
Handles: []

PGP keys: []
Wallets: []
Trust links: []
Marketplaces: ['market_beta']
Cleartext domains: []
SSL certificates: []
Certificate ↔ domain links: []
Service/default banners: []
Exposed server-status: []
Descriptor inconsistencies: []
Hidden-service identifiers: []
Persona history overlap: []
This layer is corroborating context and is not a seventh score signal.

Privacy / Anonymity Context — NOT Attribution Evidence

Original: {'tor': True, 'vpn': False, 'cryptocurrency': False, 'encrypted_messaging': False}
Candidate: {'tor': True, 'vpn': False, 'cryptocurrency': False, 'encrypted_messaging': True}
These indicators do not contribute to the attribution score.

Methodology

Temporal cosine similarity; configurable account decay; infrastructure component overlap; activity cosine similarity; interaction-network similarity; synthetic stylometry; weighted evidence fusion; separate contradiction penalty; confidence bands.

Evidence Integrity

E-caseC_original — SHA-256: 0d03ed331fc4ae1ebb8579deb11d8129fb3d2d498b50580845c2548abe8f41aa

E-caseC_candidate — SHA-256: 3d0887e3d822d596460923e9dfc648319e01437b2116cbdbb5dd2d9337e87bc1

Limitations

Controlled synthetic evaluation only. It does not establish real-world de-anonymization performance.

Decision-Support Notice: This report does not establish real-world identity or guilt.